

Manual

Tool and Resource Monitoring WRM-IHA 8.1

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Contents

1	Order Generation for Resources – Overview.....	4
2	Resource Overview	5
3	Resource history	13
4	Machine history	19
5	Maintenance Calendar (Activity Calendar)	29
6	Production Variants.....	44
7	Automatic generation of orders - processing	49
8	Generation of orders (configuration)	53

1 Order Generation for Resources – Overview

Purpose

The function package *Order Generation for Resources* provides functions to generate maintenance orders directly from the different applications of the resource management.

Integration

The function is integrated in different applications – see below. To use this function, a customer-specific configuration and a briefing of the user is required or recommended.

Features

Application service (AS) to generate maintenance orders. You can plan these orders using the HYDRA planning functions and record actual data on shop floor terminals for the orders:

- You can create maintenance orders in the maintenance calendar and define a specific order type to specifically control such orders.
- You can create general maintenance orders and edit measures in the resource history.
- You can create repair orders in the resource history.
- You can create retooling/setup orders in the HYDRA Shop Floor Scheduling if you use this tool for the detailed planning.
- You can enter the resource status when you log on, interrupt and log off the maintenance orders.
- You can create overviews and reports for maintenance orders in the HYDRA-BDE functions.
- You can perform uploads to ERP systems including the actual data recorded for maintenance orders.

Note: The precise requirements and the solutions found are implemented as part of a HYDRA configuration (chargeable service).

2 Resource Overview

Overview

Menu	Production facility management → Current information → Resource overview
Transaction code	resov
Function authorization	resov.*

The resource overview is the central application that provides an overview of the current resources managed in the system. The resource overview provides functions regarding status change, stock transfer and documentation of measures.

Purpose

You can use the *Resource overview* to show the current status of selected resources. The application shows all resources matching the conditions you specified in the selection parameters.

The list shows the specified resource master data and the current resource status. If the tool is active (logged on), the *Resource overview* shows the machine and the operation where the resource is logged on to. If the resource is locked, the overview shows the following:

- the reason of the lock,
- the time when the resource has been locked, and - if entered -
- the time until the resource will be locked.

Integration

You can use the master data function *Resource configuration* to create, change or delete resources.

Selection criteria

The application provides the following selection criteria:

Resource from ... to ...

This selection criterion refers to the resource. You can also use wildcards (placeholders *).

Short name

Short name of the resource. Only relevant for resources of type MNR.

Resource type

Type of resource.

Designation

Name of the resource.

Group

Workplace/machine group of the resource. Only relevant for resources of type MNR.

Cost center

Cost center of the resource.

Short name

Short name of the resource.

Resource family

Family the resource is assigned to.

Responsibility area

Responsibility area the resource is assigned to.

Storage location

Regular storage location of the resource.

Current storage location

Current storage location of the resource. The functions *Status change* and *Stock transfer* are used to book the resource to this storage location. If you log on a resource to a workplace, this workplace automatically becomes the current storage location of the resource.

Show deleted resources (as of WRM 8.2)

Use this field to specify if logically deleted resources are shown.

MD user fields

MD user fields 1- 6 of the resource. If you select a resource family in the selection panel, the application shows the field names according to the assigned user field definition.

Field descriptions

Most field descriptions are included in the application [Resource configuration](#). See the following tabs:

- General resource configuration
- User fields
- Comment
- Resource attributes
- Resource list

The additional tabs of the resource overview are described here:

"Status" tab**Active**

Specifies whether or not the resource is currently logged on to a workplace.

Deleted (as of WRM 8.2)

Specifies whether the resource displayed is logically deleted or not.

Current machine, current order

For active resources, this field specifies the workplace and the order where the resource is currently logged on.

Current storage location

Current storage location of the resource. The functions *Status change* and *Stock transfer* are used to book the resource to this storage location. If you log on a resource to a workplace, this workplace automatically becomes the current storage location of the resource.

Resource status, status since

Specifies the current status of the resource. Number and name are displayed and the time when the status was set.

New status, from, until

You can specify a future status that is set when the status changes. The status is not immediately set, but an expected future status is specified.

The HYDRA Shop Floor Scheduling can use this future status to show and monitor the resource assignment.

In the HYDRA Scheduler, a job "WRM resource control" is installed. This job regularly executes the script `./hyresctl.scr /STATUS` (by default every 30 minutes). The program called by the script, performs the following processing:

- If the time "New status from" is exceeded, the new status is set.
- If the time "New status until" is exceeded, then the status is set that includes the value "F" in field *Collection* in the [status configuration](#).

Technical note: Both status changes are internally performed via an asynchronous DDI call (DLG=RES_STATUS or DLG=RES_FREI).

"Use" tab**Target utilization****Cycles**

The field *Cycles* provides additional information. The cycles value defines how long the resource is to be used.

Runtime

The field *Runtime* provides additional information. It defines how long the resource is to be used.

Actual utilization

Cycles

The total number of cycles that have been recorded for the resource since initialization.

Runtime

The total run time that has been recorded for the resource since initialization.

Yield, scrap

Total of yield or scrap that has been recorded for the resource since initialization and posted to the relevant account (base quantity, primary quantity).

Run times since implementation, allocated to RPA

Account name (e.g. SUT, MUT)

Total number of hours that has been recorded for the resource and booked to this resource performance account since initialization.

Toolbar

Resource tab



File – Show file (function authorization: mdres.doc)

Opens the file view – only available for document resources, which are configured as file-based resources without DNC processing in the *Resource type*. And only available if the relevant license and function authorization are available.



Go to – Resource configuration (function authorization: mdres)

Opens the application *Resource configuration*. The selected resource is passed as default value.



Go to – Resource list (function authorization: mdrbom)

Opens the *Resource list* application. The selected resource is entered as default value for the higher-level resource.



Go to – Resource history (function authorization: reshi)

Opens the application *Resource history*. The selected resource is passed as default value.



Functions – Measure

Opens the *Measures* application.



Functions – Status change

Opens the dialog to change a resource status. The following input fields are available:

Status: Status that is set according to configuration.

Material buffer: With the status change, the resource can be relocated and transferred to another storage location.

Set immediately:

- The current date is preassigned to the field *from* and cannot be changed.
- The field *till* is empty; you can enter a date here.

Set for period:

- The current date is preassigned to the field *from* and can be changed.
- The field *till* is empty; you can enter a date here.

If no date is entered in field *till*, the system automatically sets the date to 31-DEC-9999.

The checkbox *Including subordinate resources* is not relevant and reserved for future extensions.

Comment: The comment entered can be displayed in the *Resource history*.



Functions – Release of resource

Opens the dialog to release a resource. The checkbox *Including subordinate resources* is not relevant and reserved for future extensions.



Functions – Stock transfer

Opens the dialog to transfer/relocate a resource.



Generate order (function authorization: resovgenorder)

Generates a maintenance order for the selected data row. (The function is only available with function package WRM-IHA).



Document management (function authorization: resovdoc)

Click this button to call the [Document management](#).

DNC tab

The tab is only available, if you select a DNC resource. These are resources configured with *DNC processing* in the *Resource type*.

**File – Export (function authorization: mdres.export)**

Exports the file specified for the resource. You use the file explorer to specify the target file.

**File - Import (function authorization: mdres.import)**

Imports the file specified for the resource. You use the file explorer to specify the source file.

**File - Viewer (function authorization: mdres.vis)**

Opens the file specified for the resource using the defined viewer program.

**File - Editor (function authorization: mdres.editor)**

Opens the file specified for the resource for editing using the defined editing program.

**File – Comparison editor (function authorization: mdres.diff)**

Opens the comparison editor for the selected resource or resources. See below for further information.

**Go to – Resource configuration (function authorization: mdres)**

Opens the application *Resource configuration*. The selected resource is passed as default value.

**Go to – Resource list (function authorization: mdrbom)**

Opens the *Resource list* application. The selected resource is entered as default value for the higher-level resource.

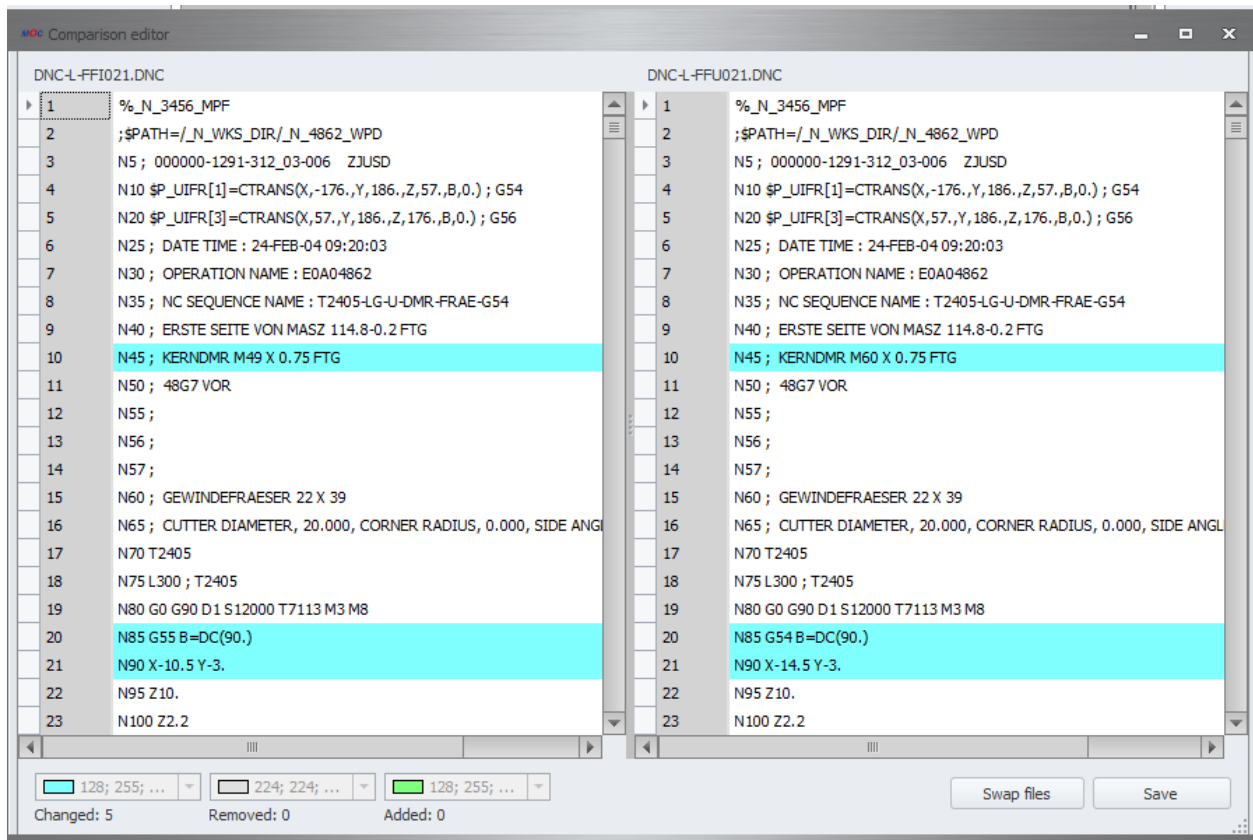
**Functions – Status change**

Opens the dialog to change a resource status.

**Functions – Release of resource**

Opens the dialog to release a resource.

How to use the comparison editor



The comparison editor compares the files attached to the DNC resources. Two operation modes are available:

Selection of one resource:

The editor shows the released resource and the optimized version of the resource for comparison. You can change the file displayed on the right-hand side of the editor. Once you have made the changes, the comparison editor transfers these changes to the system, like the simple editor. You can only use this mode for DNC types with the file processing type "optimized".

Selection of two resources:

If you select two resources before you open the comparison editor, the editor compares the two selected resources. You can select the file type. You can change the file displayed on the right-hand side of the editor. Once you have made the changes, the comparison editor transfers these changes to the system, like the simple editor.

Click the relevant buttons or use the context menu (right-click) to start the functions of the comparison editor:

- Reject: Rejects the difference identified (on the right). Accepts the value from the left file. The editor does no longer highlight the difference.
- Keep: Accepts the difference identified (on the right). The editor does no longer highlight the difference.
- Next difference: Goes to the next difference.
- Insert: Inserts a row at the current position.
- You can always change the contents of a row. Click the row and enter a value. Press ESC to quit the row without changes. The editor then highlights the row as "changed".
- Swap windows: Click this button to swap the windows. This function is necessary if you compare two resources. The place where a resource is displayed results from the display order in the table; the system does not know, which resource must be changed. In mode *Selection of one resource*, this button is not available because here you can only change the optimized program version.
- Save: Saves the changes made to the file on the left-hand side.

3 Resource history

Overview

Menu	Production facility management → Resource analysis → Resource history
Transaction code	reshi
Function authorization	reshi

The "resource history" provides an overview of what happened to a resource in the past. Therefore, you can trace activities that are relevant to quality ("log book"). The application maps the resource's "life cycle".

Purpose

In this application the system documents all actions/events relating to the object "Resource". The resource history documents the actions/events resulting from the resource management and the use of resources in BDE orders. BDE orders can be production orders and maintenance orders. The system documents the following:

- Status changes (resource status changes)
- Stock transfers
- Measures/ comments
- Exceeded maintenance cycles
- Maintenance reset
- With DNC: Upload and download
- Order logons
- Order logoffs

Whatever has been done to and with the resources is documented and available at any time. The application also includes a print function in the electronic resource book.

Selection criteria

The application provides the following selection criteria:

Resource type

This selection criterion refers to the resource type. You can also use wildcards (placeholders *).

Date from / to

Use the date selection to restrict the period of time for the data you want to evaluate.

Both times respectively refer to the start or end of the period specified above.

Resource

This selection criterion refers to the resource number. You can also use wildcards (placeholders *).

Events

Use this selection criterion to select specific events. The events you can select depend on the product groups you have licensed and the events that are actually recorded.

Workplace

Enter a workplace to select the events that have been posted in relation to this workplace. This mainly affects the following events:

- Machine status
- Beginning of status / end of status
- Production lock
- Operation postings
- Personnel postings
- Target value changes
- Resource postings
- BDE comments
- DNC upload/ download

Family

This selection criterion refers to the resource family the resource is assigned to.

Status

This selection criterion refers to the status of the resource. The system selects the events of the following types for which this resource status was set in the period entered:

- Resource status
- Release of resources
- Maintenance cycle exceeded
- Maintenance reset
- DNC upload
- DNC download

Cost center (workplace)

This selection criterion refers to the cost center stored in the machine and/or workplace master data. The application shows all machines and/or workplaces assigned to the selected cost center. You can also use wildcards.

Designation (name)

This selection criterion refers to the resource name as defined in the workplace/resource configuration.

Cost center (resource)

This selection criterion refers to the cost center of the resource as defined in the workplace/resource configuration.

Article

This selection criterion refers to the article number of the operation that was recorded with the event.

The system logs the operation with the following events:

- Logging on/off, interrupting operations (for resources of the type MNR)
- Logging on/off staff (for resources of the type MNR)
- BDE comments (for resources of the type MNR)
- Changing the target cycle (for resources of the type MNR)
- Changing the partitioning (for resources of the type MNR)
- Logging on/off resources (for resources of the type <> MNR)

MES order number

This selection criterion refers to the MES order number of the operation that was recorded with the event.

Order

This selection criterion refers to the order number of the operation that was recorded with the event.

Reporting person

This selection criterion refers to the personnel number of the employee who carried out the posting that initiated the event.

The system logs the *person* for the following events, if the person was entered with the posting:

- Logging on/off, interrupting operations (for resources of the type MNR)
- Logging on/off staff (for resources of the type MNR)
- BDE comments (for resources of the type MNR)
- Changing the workplace/machine status (for resources of the type MNR)
- Changing the target cycle (for resources of the type MNR)
- Changing the partitioning (for resources of the type MNR)
- Logging on/off resources (for resources of the type <> MNR)

Field descriptions



Subject to the logged events, the system only populates specific fields.

The "Duration" field, for example, is populated in case of a machine status change. But if the resource status changes, the system enters the value zero.

The fields are classified into the areas:

Event

Includes information on the event.

Resource

Provides information on the currently selected resources. The event usually refers to the resource mentioned in the subsection "Resource master data".



If an event only refers to a machine, the relevant machine data is entered in the "Resource master data" section and in the "Workplace master data" section.

If a DNC resource is either uploaded or downloaded, the DNC resource is entered in the "Resource master data" section. The "workplace master data" section shows the machine from which data was uploaded or downloaded.

If a resource is changed (e.g. a resource is logged on/off via a terminal), the resource is entered in the "Resource master data" section. The "Workplace master data" section includes the relevant machine.

Person

Information about the person or user performing the posting.



If you reset a maintenance activity via a terminal, the application also shows the terminal user number in the "Modified by" field. If you reset a maintenance via the MOC, the field includes the user (MOC user).

The other columns depend on the event.

Resource status

Information on the set resource status of the status event.

Order

Information on the operation that was currently processed at the time of the event (event timestamp)

Maintenance

Information on the *maintenance* event.

Measure

Information on the *measure* event

Upload/ download

Information on the DNC event *upload* and/or *download*.

Note: Two events are documented with the upload, as the upload is processed in two stages. Dialog ID "N": upload has started. "F": file has been saved in the target folder. The system only makes one entry for downloads. Only the file transfer is documented as a relevant event.

Toolbar



Resource overview

You can directly open the application "resource overview" by clicking the icon.



Generate order (reshigenorder)

Use the "generate order" function to create orders from work plans based on the specified configuration.



Document management (reshidoc)

This button is only enabled if you select a "Maintenance reset" event. The [Document management](#) opens.



Set the measure to "measure completed" (resmeasfin)

Use this function to identify an unfinished measure as being "completed". Once you have requested the function, a prompt pops up where you have to confirm your input. Once you have confirmed the dialog, the system sets the measure to "done" (column "done"/"settled": ☒) and documents the user (column "done by").



Resource book (reshi.book)

The resource book is a report providing the information of the resource history in a formatted form. The aim of this function is to make data printable. This way, the data is also available as hard copy, e.g. for QA documentation purposes. You may choose to have the data grouped by day for the daily report or by week for the weekly report.

The cover sheet shows master data information of the displayed and selected resources.

The presentation varies depending on the event.

4 Machine history

Overview

Menu	Production facility/Resource management → Resource analysis → Machine history
Transaction code	wphi
Function authorization	wphi

Purpose

The machine history is a report for the production management. The application allows for tracking and tracing of events that need to be posted at workplaces in MES. In this context, posting events such as status changes, order, tool, and personnel postings, maintenance activities as well as measures recorded at a workplace are listed in chronological order in a table. You can use various selection criteria to evaluate events.

Selection criteria

The application provides the following selection criteria:

Workplace from ... to ...

This selection criterion refers to the workplace stored in the machine or workplace master data. You can also use wildcards (placeholders *).

Group from ... to ...

This selection criterion refers to the group stored in the machine or workplace master data. The application shows all workplaces/machines assigned to the selected group. You can also use wildcards.

Short name

This selection criterion refers to the short name of machines in the master data. The application shows all machines or workplaces matching the entered character string. You can also use wildcards.

Designation

This field refers to the name of machines and workplaces defined in the machine master data. The application only shows the machines matching the specified character string. You can also use wildcards (placeholders *).

Cost center

This selection criterion refers to the cost center stored in the machine and/or workplace master data. The application shows all machines and/or workplaces assigned to the selected cost center. You can also use wildcards.

Company

This selection criterion refers to the company defined in the machine or workplace master data. The application shows all workplaces/machines assigned to the selected company. You can also use wildcards.

Report group

This selection criterion refers to the report groups. The application shows all workplaces/machines assigned to the selected evaluation/report group.

Responsibility area

This selection criterion refers to the responsibility area in the workplace/machine master. Note: The user can only view those machines included in the responsibility areas assigned to the user.

Type

Selects the category of the machine/workplace displayed in the evaluation/report. You can select E (individual workplaces) and G (group workplaces).

Type

Selects the workplace type. You can select the following workplace types:

- P Workplace
- N Machine
- J Machining center
- L Line
- A Aggregate
- C CAQ inspection station
- R Reel-based manufacturing
- S Cutting unit

Show comments

If you select the checkbox *Show comments*, the table also shows entered comments.

Comment

If the input field *Comment* includes a text, the table only shows the data records that include this text as a comment. You can use * as a wild card. Please note case sensitivity.



You cannot use this selection field to search BDE comments.

Machine statuses > X minutes only

This parameter only refers to events of the type "machine status". The application will show the machine status if the posted time is greater than the entered value.

Event type

You can restrict the displayed events. The application shows all events, in case you have not restricted the selection.

Designation	Acronym
Machine status	M_MST
Production lock	M_PSPERRE
Operation postings	A_ADE
Personnel postings	P_ADE
Target value changes	M_VORGABE
Maßnahme	R_MASSNAHME
Resource posting	R_MELDUNG
Release of resource	R_FREIGABE
Resource status	R_STATUS
Maintenance reset	R_WART_RESET
Exceeding of maintenance	R_WART_EXCEEDED
DNC Upload	R_UPLOAD
DNC Download	R_DOWNLOAD
Transfer posting of resources	R_UMBUCHUNG
Beginning of status end of status	RES_STB RES_STE
BDE comment	HY_BEM: Display of BDE comments that have been entered in reference to an operation.

Please note: Posting of events depends on the customer's system and its use. Consequently, it might be the case that not all events listed here are relevant.

Date from ...to (shift/ time)

Use the date selection to restrict the period of time for the data you want to evaluate.

When selections are made using shift(s), the shift date is evaluated. If no shift is selected, all shifts are used.



Note that selection by shift is not supported for all event types. You can find detailed information on the shift selection [here](#).

If you select by time, the selection is based on the start date. Both times refer to the beginning or end of the date period specified above.



You can only evaluate *Group workplaces* if you select by *Time*. If you select by *Shift*, no data will be displayed because group workplaces do not refer to shifts.

Order / Article / MES order number

You can use these criteria to search for BDE postings:

- Log on OP, interrupt OP, log off OP, enter part quantities
- Log on staff, log off staff
- Change partitioning, change target cycle
- BDE comment

Machine history detail application

The machine history lists all events, such as status changes, order or personnel postings of a machine that occurred on the day. These have to be evaluated or listed in a shift of this day. The evaluations/reports show the following postings:



Postings based on machines/workplaces:

Postings for machine statuses recorded automatically (with direct machine connection)

Postings assigned manually at the terminal

Setting the production lock or changing default values relating to machines/workplaces (target cycle, partitioning) at the terminal

Automatic assignment of default values with operation postings



Postings based on orders:

Postings performed automatically (when shifts change)

Manual postings (logon, logoff, interruption) at the terminal.

The corresponding order is displayed additionally. If it is a manual posting, the person who did the posting is shown as well. If waiting period processing is active, the displayed logon time of the order represents the time of entry and may deviate from the point in time indicated in the order log record. →→→

**Postings based on staff:**

Automatic (when shifts change)

Manual logon or logoff processes of staff at the terminal

In addition, the application shows the corresponding personnel number and the operation for which the person produces.

**Postings based on resources:**

Machine postings resulting from the HYDRA Tool and Resource Management module (HYDRA-WRM), e.g. the application also shows exceeded maintenance activities or measures/comments.

**Information**

Shows BDE comments entered via the AIP terminal and stored with the operation.

The event "information" also shows the total duration of the respective status / event. The duration is always zero when a person or OP is logged on. The duration states the interval between the logging on and logging off if you interrupt/log off an OP or person.

Field description

The following paragraphs describe the data available in the table. It might be the case that the application does not show this data by default. Use the column selection function to add the required data.

Field description *workplace* category

Workplace

Workplace the event refers to.

Field description *event* category

Type

Image display of the type

Event type

Assign the recorded event. Possible values: see event

Event

Classifies the event collected at the machine in the table row. In the columns "Selection by shift" and "Selection by time" you can see events available for a specific selection.

Event type	Event	Selection by shift	Selection by time
Machine status	Machine status according to configuration Coloring is set according to the settings in the status text	Yes	Yes

Event type	Event	Selection by shift	Selection by time
	configuration..		
Production lock	Production lock set manually Production lock canceled manually	No	Yes
Operation postings	OP logged on OP interrupted OP logged off	Yes	Yes
Personnel postings	Person logged on Person logged off	Yes	Yes
Target value changes	Change partitioning/change target cycle	No	Yes
Exceeding maintenance of	Maintenance cycle exceeded	No	Yes
Maintenance reset	Maintenance reset	No	Yes
Information	BDE comment entered	Yes	Yes
Beginning of status end of status	Event and coloring according to configuration	No	Yes

Datum

Entry date of the event

Time

Entry time of the event

Duration

Time between the last event of this kind and the one currently displayed. The duration is only shown for the events "OP INTERRUPTED", "OP LOGGED OFF", "PERSON LOGGED OFF" as well as for machine statuses. In any other case, 0 is shown. These durations are synchronized with the BDE shift calendar, i.e. shift breaks are not included. Consequently, this value does not necessarily correspond to the period of time between logon and logoff.

Field description *master data* category

Workplace

Unique ID defined in the workplace configuration.

Designation

Machine name as defined in the workplace configuration.

Comment

Comment on the machine as defined in the workplace configuration.

Group

Capacity group which the machine was assigned to.

Cost center

Cost center as defined in the workplace configuration.

Company

Company as defined in the workplace configuration.

Responsibility area

Responsibility area required to view this workplace as defined in the workplace configuration.

Field description *order* category**Order type**

Order type of the order for which the event was collected.

order

Order number of the OP for which the event was recorded.

Sequence

Sequence number of the OP (provided that sequences are used).

OP

Operation number

Split

Split number of the operation (if split OPs are used)

SOP

Sub operation number (reserved).

Article

Article number produced by the operation; taken over from operation data.

Article designation/name

Article name of the article.

Field description *person* category**Person**

Personnel number of the person that has been logged on or off (only for Pers. postings)

Last name

The person's last name who was logged on or off (for personnel postings only).

First name

The person's first name who was logged on or off (for personnel postings only).

Name

Full name (last name, middle name and first name) of the person who was logged on or off (for personnel postings only).

Field description *status* category

If the event is a machine status, then this category shows the status number and status text name. This category shows the resource status for events based on resources.

Status

Status number of the assigned status

Status text

Status text of the assigned status

Receiving storage location

Destination when entering a resource status change (RES_STATUS).

Field description *maintenance* category

Maintenance type

Type of the maintenance

T: based on cycles,

B: based on operating hours

Z: based on time

Maintenance

Maintenance short text

Target cycles

For maintenance type T only: number of cycles until the maintenance is due again.

Actual cycles

For maintenance type T only: number of cycles accrued since resetting the maintenance interval. Value results from the machine data collection (MDE).

Planned hours of operation

For maintenance type B only: number of operating hours until maintenance falls due again.

Actual hours of operation

For maintenance type B only: number of operating hours accrued since resetting the maintenance interval. Value results from the machine data collection (MDE).

Next date

For maintenance type Z only: time when the maintenance falls due the next time.

Processing mode

For maintenance events (RES_WART):

R = Reset

Z = Threshold exceeded

A = Enabled/disabled

For changed resource statuses (RES_STATUS):

S = Change over status

Threshold 1 (in %)

Threshold until reaching due date

Threshold 2 (in %)

Threshold until reaching due date

Threshold 3 (in %)

Threshold until reaching due date

Active

“Active” flag of the maintenance activity at the time of the event.

Active (so far)

Only relevant for processing mode A: previous “active” status of the maintenance activity at the time when the maintenance activity was activated/deactivated.

Modified by

Editor who edited/set/reset the maintenance.

Datum

Date of editing/resetting

Time

Time of editing/resetting

Field description *measure* category

Maßnahme

Measure name

Designation

Name/description (long text) of the measure.

Reporting person

Person who created the measure.

Verantwortlicher

Person who has to carry out the measure.

Date of solution

Date when the measure has to be completed.

Priority

Priority of the measure.

Done

Flag indicating that the measure has been completed.

Done by

Person who marked the measure as being completed.

Field description *upload/download* category

(Not supported)

Field description *comment* category**Comment**

Comment on the event entered by the employee.

Field description *changed partitioning* category**Partitioning**

Partitioning

Cavity

Cavity number.

Type of modification

Reduced partitioning or increased partitioning.

Reason for change

Number of the reason for change.

Text of reason for change

Text of reason for change

Toolbar

Generate order (function authorization wphigenorder)

Use the "Generate order" function to create orders from work plans based on [Configuration](#).



Order information (function authorization: orin)

Request [Order information](#).

5 Maintenance Calendar (Activity Calendar)

Overview

Menu	Resource Management → Current information → Maintenance calendar
Transaction code	rmcal
Function authorization	rmcal

This overview is a valuable aid for supervisors and maintenance personnel because only through regular maintenance, the expensive resources (e.g. production machines or tools) can retain their production quality and do not cause unnecessary breakdowns. You can also use the calendar as basis for maintenance planning and as data collection tool to record the activities performed.

Purpose

The maintenance calendar or activity calendar has been designed to plan and show maintenance activities or other recurring activities. Activities can be maintenance, test equipment calibration, reading of energy counters and so on. Use the field *Activity type* to identify the relevant function and type of activity. In most cases, this field remains empty which means that a maintenance or similar activity is scheduled for a resource. For special requirements, e.g. calibration of test equipment, enter the relevant identifier in this field ("K" to identify calibration of test equipment). That means, the user can differentiate between the type of activity.

This document describes how to use the calendar in the HYDRA Tool and Resource Management, Energy Management and Gage Calibration.

The task of maintenance and activity monitoring is to track the configured activities and perform the following actions:

Refreshing the current values

- "Cycles recorded so far" (cycle-based maintenance/activity) or
- "Hours recorded so far" (maintenance/activity via hours of operation).

The status of an activity is set when a configured threshold value (blue/yellow/red) has been exceeded, and this event is documented in the database (to generate evaluations via the resource history). The threshold values are checked in the following order: red > yellow > blue. This means, the system checks first if the threshold value "red" has been exceeded. If so, this status is set and documented. Otherwise, the inspection is continued for the threshold values "yellow" and then "blue".

Monitoring is only run for activities that are marked active and whose validity period includes the current time.

For this purpose, the monitoring process `hywtcupd.out/.exe` is embedded in the HYDRA scheduler and cyclically called. .

You can define any number of activities for each resource. Several activities can thereby be defined for each resource. The following types of intervals are available when defining the maintenance times:

Cycle-based activity

The system compares target and actual cycles with a cycle-based activity. The difference between the two values indicates when maintenance is due. The actual cycles are automatically recorded in HYDRA.

Requirement: a cycle monitoring must be performed for the machine.

Activity based on hours of operation

The times recorded in HYDRA are used for a maintenance/activity based on hours of operation. In the *Resource type*, you specify the resource performance accounts that are used to calculate the hours of operation. The activity is due when the interval defined in the maintenance calendar has been reached.

Time-based activity (days)

With this type of activity, the system calculates the next maintenance due date using the number of days specified for this activity. This number of days is based on the Gregorian calendar.

Single activity

Combined with the above-mentioned types of intervals, you can even specify an activity as "non-recurring". After the reset, the activity is deactivated automatically.

Additional notes:

When data is selected, the user can only view the activities of resources that are included in the responsibility area the user is authorized for.

You must have defined the resource master data in the resource stock, which is required for the maintenance calendar.

To use resources with cycle-based activities and maintenances based on hours of operation, it must be configured in the system that you can post to these resources. This can be configured via the resource type.

For resources with DNC processing, no cycle-based activities and no activities based on hours of operation can be defined, as you cannot post data for these resources.

And also for energy counters this is not possible, as energy counters do not use machine cycles as data basis.

Selection criteria

The application provides the following selection criteria:

Resource type

Selection of the specified resource type.

Resource

Selection of the specified resource.

Field description of the *Activity* tab

Resource type

Shows the resource type for the defined the activity.

To use resources with cycle-based activities and maintenances based on hours of operation, it must be configured in the system that you can post to these resources. This can be configured via the resource type.

For resource types with DNC processing, no cycle-based maintenance and no maintenance based on hours of operation can be defined, as you cannot post data for these resources.

Resource

Shows the resource the activity is defined for.

Activity

Description of the activity. When you create an activity, the system automatically allocates a number to the name and this number identifies the activity.

Type

Select a type to specify how monitoring is carried out:

- T Cycle-based activity
- B Activity based on hours of operation
- Z Time-based activity

Depending on the above selection, one of the tabs *Cycles*, *Hours* or *Days* is released.

Single activity

If this option is set, the activity is only carried out once. In this case, the "interval" field is hidden. When reset, the activity is automatically deactivated.

Class

This input field is used to classify maintenance activities. For example, all cleaning activities can be classified as "Cleaning".

Using the grouping function in the overview screen, you can combine and display all maintenance activities that logically belong to the same class. Otherwise, this entry field is used for comments.

Active

It is possible to deactivate activities temporarily, and reactivate them again at a later time. The following display shows if an activity is currently active or not:

-  Activity activated
-  Activity deactivated

Deactivated activities are not integrated in the monitoring.

Authorization (as of service pack 16/2020)

You can use the authorization level to specify the maintenances in detail that a person is allowed to reset on the Windows terminal AIP.

A person can reset a maintenance if the following conditions are fulfilled:

- The option (checkbox) *Reset maintenances* in the HR master data is activated;
- An authorization level is entered in the HR master data;
- The authorization level of the person must be greater than or equal to the authorization level specified in field *Authorization* of the maintenance.

The system performs an online validation check on the HYDRA server. The system only performs the authorization check if a staff badge number has been entered in the AIP input dialog.

If no authorization level is specified, then the maintenance can always be reset and the authorization level stored in the HR master data is not relevant (downward compatibility).

Status

A colored signal in front of each activity clearly shows if a maintenance interval will soon expire or has already expired, and if a maintenance activity must be carried out. This way, the user can see at one glance which activity must be carried out or is already overdue.

The user specifies threshold values, which change the color of the signal. For cycle-based activities and maintenances based on hours of operation, the application shows percentages, for time-based activities days are shown. The following 4 colors can be defined (each corresponding to a status type):

-  green
-  blue
-  yellow
-  red

For further information, please refer to the descriptions of the different interval types (see below).

Note:

The status of the resource itself does not change when a maintenance status is reached.

Last activity

This shows the time when the maintenance activity was last reset and the name of the user who reset the maintenance.

Please note that the time of resetting the maintenance may deviate from the time that the maintenance has actually been carried out.

Valid from, valid until

You specify these times to assign time limits to maintenance activities.

If the current point in time is not included in the validity period of a maintenance activity, then this maintenance is not integrated in the maintenance monitoring, i.e. the maintenance status is not updated.

Modified by

Name of the last user who edited the maintenance activity and time of the last change.

The different tabs are explained in the following:

Field descriptions of the *Cycles* sub-tab

Interval

After the number of machine cycles specified in this field, the maintenance must be carried out. This number and the two values below refer to the value in the *Reference* field (see below).

This field is hidden if it is a singular/non-recurring maintenance.

Cycles recorded so far

Here, the system displays the number of machine cycles recorded so far in HYDRA. This value is updated by a cyclical process. For further information, refer to the section *Maintenance monitoring*.

Next activity

If you create a new activity, then this value is calculated by default using the current actual value (actual cycles) + the specified interval.

When you reset the maintenance activity, the system calculates this value and shows when the next activity is due.

When an activity is reset, you can use the *Calculation base* option to specify how the next maintenance due date is calculated:

Target value The value for the next activity is based on the current value:
Next activity after = current value (next activity after) plus interval

Actual value The value for the next activity is based on the number of previously recorded cycles:
Next activity after = cycles recorded so far + interval

Reference

For the above values, you must always consider this option. The following values are possible:

G Total
Activity monitoring is based on the total number of cycles recorded so far.

A Relating to order/OP
If a resource is logged on with the operation logon, then it is checked if activities with reference = A exist for this resource. These activities are then automatically reset.

The *monitoring* now checks whether the interval has been reached using cycles posted for the operation currently logged on and sets the status accordingly.



Order-related monitoring is not available for resources of the type "MNR" (machines)!!

This type of maintenance monitoring only makes sense for resources that are logged on to one operation at a given time. This means that a maximum of one operation may be logged

on to the workstation/machine.

The posting of cycles to a resource does not take place in real time, but at longer intervals (e.g. at logoff, at interruption of an operation or during an automatic change of shifts).

Therefore, this type of monitoring only makes sense for operations with a long runtime.

Blue / Yellow / Red

Enter the threshold values as percentages that identify the status of a maintenance activity.

"Cycles recorded so far" < Blue-% from "Next maintenance after"

"Cycles recorded so far" >= Blue-% and < Yellow-% from "Next maintenance after"

"Cycles recorded so far" >= Yellow-% and < Red-% from "Next maintenance after"

"Cycles recorded so far" = Red-% from "Next maintenance after"

The relevant values specify the signal color.



Notes



The threshold values can be greater than 100%.

No validation check is made with regard to the order of the threshold values.

Field descriptions of the *Hours* sub-tab

Interval

Enter the period of time after which the maintenance activity must be run. This value, and the two following values, refer to the value in the *Reference* field (see below).

This field is hidden if this is a single activity.

Hours recorded so far

Here, HYDRA displays the time that has been posted for this resource so far. This value is updated by a cyclical process.

Please note that for the hours recorded so far, the RPA times are used that are identified as such in the *Resource type* (option *RPAs as hours of operation in the Maintenance calendar*).

Next activity after

This value is calculated by default from the current actual value (hours recorded so far) plus the specified interval.

When the activity is reset, this value is calculated and shows when the next maintenance activity is due.

When a maintenance activity is reset, you can specify how the next activity is to be calculated using the "Calculation base" option:

Target value The value for the next activity is based on the current value:
 Next activity after = current value (next activity after) plus interval

Actual value The value for the next activity is based on the number of hours recorded so far:
 Next activity after = hours recorded so far + interval

Reference

For the above values, you must always consider this option. The following values are possible:

- G Total
 Activity monitoring is based on the total time that has been posted so far.
- A Relating to order/OP
 If a resource is logged on with the operation logon, then it is checked if activities with reference = A exist for this resource. These activities are then automatically reset.

On the basis of the duration posted for the currently logged on operation, monitoring now checks whether the interval has been reached, and then sets the status accordingly.



This type of maintenance monitoring only makes sense for resources that are logged on to one operation at a given time. This means that a maximum of one operation may be logged on to the workstation/machine.

The posting of cycles to a resource does not take place in real time, but at longer intervals (e.g. at logoff, at interruption of an operation or during an automatic change of shifts). Therefore, this type of monitoring only makes sense for operations with a long runtime.

Blue / Yellow / Red

Enter the threshold values as percentages that identify the status of a maintenance activity.

"Hours recorded so far" < Blue-% from "Next activity after"

"Hours recorded so far" >= Blue-% and < Yellow-% from "Next activity after"

"Hours recorded so far" >= Yellow-% and < Red-% from "Next activity after"

"Hours recorded so far" >= Red-% from "Next activity after"

The relevant values specify the signal color.



green



blue



yellow



red



The threshold values can be greater than 100%.

No validation check is made with regard to the order of the threshold values.

Field descriptions of the sub-tab Days

Interval

Interval in days after which an activity is to be performed. The interval is based on the Gregorian calendar.

This field is hidden if this is a single activity.

Next activity on

Date when the next activity is due.

When a new maintenance is created, this value is calculated by default from the current date plus the specified interval.

Blue / Yellow / Red

Enter the number of days that specifies the status of a maintenance activity. The color of the signal is based on the remaining time, i.e. the difference between the date of the next activity and the current date ("today").

Remaining time <= "Red" value	 red
Remaining time <= "Yellow" value	 yellow
Remaining time <= "Blue" value	 blue
Other	 green



The threshold values can be greater than 100%.

No validation check is made with regard to the order of the threshold values.

Field description of the *Assignment* tab

Order

This field is only relevant in connection with the additional feature *Generate maintenance orders* or if you generate calibration (inspection) orders. Assign a maintenance order/calibration order using

the "Create order" function that you call with this button



If this field is filled, then the order number refers to a maintenance/calibration order. The activity will automatically be reset if the maintenance/calibration order is finished. When the maintenance/calibration order is finished for this activity, the order number is also removed from this input field.

Project number

This field is only relevant with activity type "K" (calibration). Depending on the system configuration, two different variants exist.

Variant 1 (there is exactly one work plan for all calibration inspection plans):

=> Input of the calibration inspection plan number (without version number)

Variant 2 (there is a separate work plan for each calibration inspection plan)

=> should remain empty. If you fill the field, the work plan list called in the application Generation of orders is pre-filtered by this project number.

Planned order

Control field that is currently not used. Remains empty.

Cost object

Control field that is currently not used. Remains empty.

Activity type

Identifies the activity type: For example, calibrations are identified by the type K.

Field descriptions of the *Information* tab

You can store a short description of the maintenance activity to ensure that the user or the maintenance worker receive more details on running the activity (e.g. notes on regulations to be observed, materials to be used).

Field descriptions of the *Resource information* tab**Inventory number**

Shows the inventory number stored in the resource configuration. Additional information in form of comments.

Engraving number

Shows the engraving number on the device (machine, radiator etc.) stored in the resource configuration. Additional information in form of comments.

Drawing number

Shows the drawing number stored in the resource configuration. Additional information in form of comments.

Manufacturer

Shows the drawing number stored in the resource configuration. Additional information in form of comments.

Owner

Shows the owner name stored in the resource configuration. Additional information in form of comments.

Toolbar**Activate**

Function authorization: rmc.al.active

Opens the editing dialog to activate an activity

**Deactivate**

Function authorization: rmc.al.deactive

Opens the editing dialog to deactivate an active activity

**Monitoring**

Function authorization: rmc.al.monitor

Updates the status of the activities

**Reset**

Function authorization: rmc.al.reset

Opens an editing dialog to reset an activity

**Capture reading (EMG 8.1 only)**

Function authorization: rmc.al.captvalues

Opens the editing dialog to enter a counter reading. Any number of difference values may be entered within a time interval. Therefore, it is a delta collection. For this reason, it is also possible to subsequently enter data relating to the past. If you do not enter date values in the fields, the system generates the interval using the last data capture as start time and the current posting time as the end of the interval.

**Enter absolute value (EMG 8.1 only)**

Function authorization: rmc.al.captvalues

Opens the editing dialog to enter a counter reading using an absolute value. The system calculates the difference to the previous reading. It is not possible to enter values for periods in the past. The system sets the start time of the interval to the end of the last entry. If an end is not entered, the system uses the current posting time.

**Generate order**

Function authorization: rmc.al.generate

Creates an order that is used for organizational processing of the activity. Once the order is finished, you can set an option to have the activity automatically reset.

As of service pack 13/2018, you can generate orders automatically. Please find further details in section "[Automatic generation of orders](#)".

**Activity plan (EMG 8.1 only)**

Function authorization: rmcaltimetable

Opens the report [Maintenance plan](#).

**Document management (WRM-WWR 8.2 only)**

Function authorization: rmcaldoc

Click this button to call the [Document management](#).

Detail application *Resources logged on*

For resources of type "MNR", the detail application provides additional information on the resources currently logged on.

Resource type, resource

The currently selected resource and its resource type.

Resource, family, resource type

Resource that is logged on, the family and the resource type.

Login

Date and time of the resource login.

Advance logon

Identifier of the advance logon.

Automatic generation of orders

When a threshold value (blue, yellow or red) is reached for an entry in the activity calendar, an associated order can be generated automatically. In the INI configuration, you define the resource type and the threshold value to generate the order automatically. If the threshold value is exceeded, then the system does not generate an order for the exceeded threshold value. If no relevant INI configuration exists, then the automatic order generation is omitted.

INI configuration

Create the INI configuration with the name „MAINTENANCECALENDAR“ and the MOC user "0". Create a new entry in the [INI configuration](#):

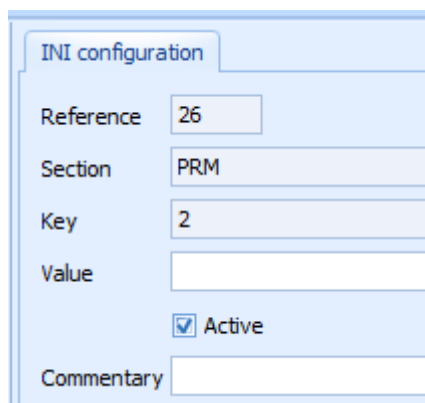
Field name	Value
Name	MAINTENANCECALENDAR
Description	Configuration maintenance/activity calendar

Field name	Value
MOC user	0

Create the required configurations for this INI configuration. For this entry, create an entry including the following values in the [INI data configuration](#):

Field name	Value	Note
Section	<Resource type>	e.g. PRM (for test equipment)
Key	<threshold value>	Threshold value blue: 1 Threshold value yellow: 2 Threshold value read: 3
Value		Leave field empty
Active	Yes	

The below screenshot outlines a configuration to generate a calibration order for the resource type "PRM" (test equipment) when the threshold value 2 "yellow" is reached.



The screenshot shows a software window titled "INI configuration". It contains several input fields and a checkbox. The "Reference" field has the value "26". The "Section" field has the value "PRM". The "Key" field has the value "2". The "Value" field is empty. There is a checked checkbox labeled "Active". At the bottom, there is a "Commentary" field which is also empty.

Order generation

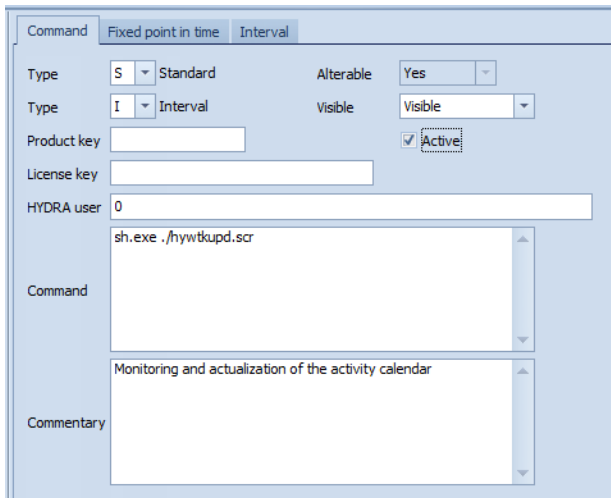
For automatic order generation it is assumed that for each resource type defined in the INI configuration in the MOC application (transaction code "edworgen" - separate licensing required) a corresponding configuration with matching object and assigned work plan is defined.

For automatic order generation it is assumed that for each resource type defined in the INI configuration in the MOC application „[Hyperlink](#)“ (transaction code "edworgen" - separate licensing required) a corresponding configuration with matching object and assigned work plan is defined.

..\..\functions\MOC\MOC_OrderAutomaticGeneration.pdf

Scheduler job

Another requirement is that the scheduler job hywtcupd.scr is active that is responsible for the maintenance status.



Create calibration order

To create a calibration order, the system also requires that the calibration inspection plan is stored in the field *Project number* of the activity calendar.



The program, which is called by the scheduler job, only generates an order if a configured threshold is actually exceeded. If a threshold has already been exceeded and the scheduler job is started again, then the program does not generate an order for this threshold.

The program also does not generate an order if the threshold 2 is configured, for example, and it is now changed from threshold 1 directly to threshold 3 because of the actual data.

Report *Maintenance plan* (EMG 8.1 only)

The report *Maintenance plan* shows all maintenance activities that are displayed in the calling application *Activity calendar*.

The maintenance activities are displayed in the following sequence:

- descending, sorted by state
- ascending, sorted by priority

The following data is displayed for each maintenance activity in the maintenance plan:

Left-hand column

- State (0: green, 1 blue, 2 yellow, 3 red)
- Maintenance type

- Description
- Resource type
- Resource

Central column

- Next maintenance after (cycles)
- Cycles recorded so far
- Next maintenance after (hours)
- Hours recorded so far
- Next maintenance on

Right-hand column

- Information

6 Production Variants

Overview

Menu	Master data → Production control → Production variants
Transaction code	prodvar
Function authorization	prodvar

Purpose

You can use the function to create or modify production variants in the system.

Integration

You can use production variants to define different options for specific articles. The system integrates these options during planning in HYDRA Shop Floor Scheduling (HLS). You can define different production variants for one article/item. An alternative production variant must exist for the article, if the order type specifies that production variants are used for planning. Otherwise, you cannot plan the operation for the machine.

If you do not want to use production variants for all areas, they can be "skipped" for single groups. Consequently, the [Group](#) configuration takes priority over the order type configuration.

An exceptional case occurs if "N" - no utilization" is configured for the order type. In this case, production variants are generally not used for this order type. The group configuration does not have any effect in this case.

Selection criteria

The application provides the following selection criteria:

Article

The system searches for production variants available for the entered article/item.

Workplace

Number of the workplace / machine where you can manufacture the article with the resource.

Workplace group

Number of the capacity group where you can manufacture the article.

Resource

Resource (number) you can use to manufacture an article as part of a production variant.

If you want to include a resource (e.g. tool) in checking for valid production variants and in planning, you should specify the resource here, otherwise leave the field empty. You cannot define a production variant at the same time for both a resource and a resource family.

Resource family

The system does not support this field when identifying production variants. You can merely use it for comments.

Status (blocked/released)

Possible values describing the current status of a production variant.

Only one version at a time may have a status value of *Released* per production variant. You can use this field to filter by released or blocked production variants.

Detail application "production variants"

The table view of the detail application offers an overview of existing entries.

Field description

Article

Article number of the article to be manufactured.

The article number length is restricted to 15 characters if HYDRA ALS (interfacing to the ARBURG host computer system) is used.

Article designation/name

Name of the article.

Group

Number of the workplace group where the article can be manufactured with the tool.

Complete this field in any case, even if the production variant should only be based on the combination of *Article* and **Workplace**.

Workplace

Number of the workplace / machine where you can manufacture the article with the resource.

Leave this field empty, if the production variant should only be based on the combination of *Article* and **Group**.

Resource type

Type of the resource you can use to manufacture an article as part of a production variant. The system only supports the resource type "WNR".

Resource

Resource (number) you can use to manufacture an article as part of a production variant.

If you want to include a resource (e.g. tool) in checking for valid production variants and in planning, you should specify the resource here, otherwise leave the field empty. You cannot define a production variant at the same time for both a resource and a resource family.

This is a key field if the interfacing to the ARBURG host computer system (ALS) is used. The resource number length (tool number) is restricted to 15 characters in this case.

Resource family

The system does not support this field when identifying production variants. You can merely use it for comments.

Please observe the following restrictions:

- If you create a new production variant and assign a resource, the resource family is not automatically taken from the pool of resources.
- The system does not check whether the resource family is valid. In case of doubt, you can select the resource family via the search dialog.
- If you change a resource family for a resource in the pool of resources, the system does not synchronize this with the production variant.

Number of resources

The number of resources required for the production variant.

Machine/operator relation for setup/production including qualifications

Define the personnel requirements that are needed for setting up or producing the operation including the relevant qualification.

You cannot use a production variant to reset (set to zero) the workforce requirements previously defined for the OP.



You can define the machine/operator relation for production variants as of SP13 and HLS-FFV 8.2. Existing customers who want to use the fields must execute the database patch provided by SP13.

Priority

Priority for identifying the production variant. A higher value stands for a higher priority.

Version

You can use this field (as part of the key) to assign version numbers to production variants. Enter "1" at the very left of the field if versioning is not used.

Only one version of a production variant at a time may have the "released" status (cf. *Status* field description).

Target cycle

Target cycle for machine monitoring in MDE. This value is defined per 1000 machine cycles.

The system keeps the target cycle defined for the operation, if you enter 0 in the "target cycle" field.

Admissible deviation

Reserved.

Partitioning

Number of the quantity produced during one machine cycle. For each machine cycle, the system enters/posts a quantity produced that corresponds to the partitioning value.

The system keeps the partitioning defined for the operation, if you enter 0 in the "partitioning" field.

Setup time

When planning in HYDRA Shop Floor Scheduling, the system uses this value instead of the default setup time value defined for the operation.

Please note: If setup time is stored as a formula for the operation, the setup time value is recalculated each time you change the group manually or you update the order. In this way, the setup time value entered here can be overwritten again.

Teardown/retooling time

When planning in HYDRA Shop Floor Scheduling, the system uses this value instead of the default teardown/retooling time value defined for the operation.

Please note: If retooling/teardown time is stored as a formula for the operation, the retooling/teardown time value is recalculated each time you change the group manually or you update the order. In this way, the retooling/teardown time value entered here can be overwritten again.

Valid from

Date from which the production variant is valid.

Valid until

Date up to which the production variant is valid. Enter 31.12.9999 here if the production variant should be valid until further notice. The system does not update the "valid until" date if the status is changed.

Status

Possible values describing the current status of a production variant:

F	Released
S	Blocked

Only one version at a time may have a status value of *Released* per production variant. If you create a new production variant with the *released* status or if you change the status of an existing variant to *released*, HYDRA checks whether an existing production variant has already the *released* status. If so, the existing version is automatically set to the *blocked* status.

Blocking reason

You can enter a numeric blocking reason here if a production variant is set to the "blocked" status.

Comment

Comment field for this production variant.

Toolbar**Generate order**

Use the "generate order" function to create orders from work plans based on the specified configuration.

Activation of production variants

You can configure in the order type if you want to use production variants. Select the order type and modify the entry for "consideration of production variants in planning".

Consideration of production variants in planning:

- Identification (E): you can select from existing production variants.
- Checking only (P): the system only checks whether a valid production variant exists.
- No use (N): the system does not check the production variants.

7 Automatic generation of orders - processing

Overview

	→	
Menu	Production facility (resource) management → Current information → Maintenance calendar → button "Generate order"	
Function authorization	rmcalgenorder	
Menu	Production facility (resource) management → Current information → Resource overview → button "Generate order"	
Function authorization	-	
Menu	Production facility (resource) management → Resource analysis → Resource history → button "Generate order"	
Function authorization	reshigenorder	
Menu	Production facility (resource) management → Resource analysis → Machine history → button "Generate order"	
Function authorization	wphigenorder	

Purpose

You can call the function "Generate order" from different MOC applications. Use this function to generate an order from a work plan.

Integration

You can use one of the following applications to create an order:

- Maintenance calendar
- Resource overview
- Resource history
- Machine history
- Production variants



For this purpose, the toolbar of these applications provides the button "Generate order".

Requirements

You need a license to call the function "generate order".

You also have to configure the following in the MOC:

- Correct configuration of [Number ranges](#):
 - o Object: order number
 - o Key: order type
 - o Value: 6
 - o Type: automatic number assignment
 - o Prefix: IH *)
 - o Ranging from: 100000 *)
 - o Ranging to: 999999 *)

*) Assumption: 8-digit order number length.

- The work plans used according to the configuration must exist
(By default: work plan IH100000)

Generating orders

Proceed as described below to execute the function "generate order":

- Select a resource / data record.
- Click the function "generate order".
- Enter the data requested in the respective dialog.

Once you have entered and confirmed the data, the system transfers the data entered in the input fields (including invisible fields) to the service. The order is generated based on the transferred work plan.

Result of order generation

The system generates the order.

If order generation was successful, the application *Edit orders* opens automatically. The input field *Order* includes the new order number and data is requested automatically.

Note:

If the *Additional data* tab does not include the order number (or the dialog has been customized and the field is no longer available), you have to configure the automatic number assignment since no target order number is indicated.

The system issues an error message (e.g. order number is missing) if the order could not be generated.

Configuration

By default, work plan number "IH100000" (the number of zeros matches the order number length defined in the basic settings) is predefined for all maintenance orders. The work plan number is configured in the MOC application [Order generation](#). If necessary, you can change this number according to your requirements. To do so, you need the corresponding license.

You can change the work plans, if you have purchased the corresponding license.

Default dialogs

Maintenance calendar

Field	Default assignment
Resource type	Resource type
Resource	Resource
OP name/designation	Description of the activity
Basic start date	Current date
Basic end date	(none)
Order	Work plan number The system takes the number of the work plan that should be used to generate an order from the configuration " Order generation ". You have to select a work plan if the configuration does not include a work plan.
Final article	Resource
Transfer as production resources and tools	Resource type = MNR No default assignment Resource type <> MNR Checked by default " ☒ ". In this case, the resource is also created as production resource and tool.

Resource overview

Like the maintenance calendar, except for the OP name. Here, the OP name is "maintenance order".

Resource history, machine history

Like the maintenance calendar, except for the OP name. Here, the OP name is "maintenance order".



You can use the MES Development Suite to configure the default assignments in the MOC application dialogs where you started the *Generate order* function. You need a license and training in order to use the MES Development Suite.

8 Generation of orders (configuration)

Menu	
Function authorization	edworgen

Purpose

You can call the function "[Generate order](#)" from different applications. Using this function you can generate an order from a work plan.

You use this application to make the required configuration for the workplace assignment.

Integration

In specific applications, you can generate a maintenance order. For these orders, the function "[Generate order](#)" uses the configurations defined here.

Configuration

You must configure the following values to generate the order:

Module

Key/name of the application, which calls the function. You can call the function "Generate order" from the following applications:

- MaintenanceCalendar → Maintenance/Activity calendar
- WorkplaceHistory → Machine history
- ResourceHistory → Resource history
- OverviewResources → Resource overview
- ProductionVariants → Production variants.

Category

Definition of the type (see configuration of the field "type") that the calling dialog uses to call the function "Generate order". The parameters of the selected row in the calling function specify the category.

Examples:

Module (application)	Category	Type
MaintenanceCalendar	maintenance.resource.type	MNRMNR
MaintenanceCalendar	maintenance.resource.type	WNR
MaintenanceCalendar	maintenance.resource.type	DEFAULT
OverviewResources	resource.type	DEFAULT
ResourceHistory	resource.type	DEFAULT

Module (application)	Category	Type
ProductionVariants	resource.type	DEFAULT

Type

Value for the configured category.

This is the resource type for which you want to provide a work plan. DEFAULT is used, if there is no specific entry for a resource type.

Work plan

Number of the work plan: The order is generated from the work plan specified here.

Object

Object that you want to call. The name is the internal application name of the function "Generate order".

Requirement for the configuration / to call the application

Activate the HYDRA Professional Mode to be able to make configurations.

Configuration procedure / assignment of the work plans

Use the column "Module" to specify the source application that is used to generate the order. The column "Type" defines the resource type for which the entry is valid. The field "Work plan" specifies the work plan that should be used.

Example 1:

Module = MaintenanceCalendar

Type = WNR

Work plan = IH100001

➔ If a maintenance for a resource of resource type WNR is selected in the maintenance calendar and if you want to generate an order automatically, then the work plan IH100001 is used to generate the order.